Augmenting a Graph of Minimum Degree 2 to have Two Disjoint Total Dominating Sets

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Abstract

A total dominating set of a graph is a set of vertices such that every vertex is adjacent to a vertex in the set. We show that given a graph of order n with minimum degree at least 2, one can add at most $(n-2\sqrt{n})/4+O(\log n)$ edges such that the resulting graph has two disjoint total dominating sets, and this bound is best possible.

Key words: graph augmentation, total domination, domatic number

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1 Introduction

A classical result in domination theory is that if S is a minimal dominating set of a graph G without isolates, then $V - S$ is also a dominating set of G. Thus, the vertex set of every graph without any isolates can be partitioned into two dominating sets.

However, it is not the case that the vertex set of every graph can be partitioned into two total dominating sets, even if every vertex has degree at least 2. (Recall that a total dominating set is a set S such that every vertex in the graph is adjacent to some vertex of S; see [4].) For example, the vertex set of C_5 cannot be partitioned into two total dominating sets. A partition into two

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total dominating sets can also be thought of as a 2-coloring of the graph such that no vertex has a monochromatic (open) neighborhood.

Zelinka [6,7] showed that no minimum degree is sufficient to guarantee the existence of two total dominating sets. In contrast, results of Calkin and Dankelmann [2] and Feige et al. [3] show that if the maximum degree is not too large relative to the minimum degree, then sufficiently large minimum degree does suffice. Heggernes and Telle [5] showed that the decision problem to decide if there is a partition of V into two total dominating sets is NP-complete, even for bipartite graphs.

In this paper we consider the question of how many edges must be added to the graph G to ensure the partition of V into two total dominating sets in the resulting graph. We denote this minimum number by $td(G)$. It is clear that $td(G)$ can only exist for graphs with at least four vertices (we do not allow loops).

The problem for paths, cycles and trees was investigated in [1]. In particular, it was shown that if T is a tree with ℓ leaves, then $\ell/2 \leq td(T) \leq \ell/2+1$.

We show that if G is a graph of order $n \geq 4$ and minimum degree at least 2, then $td(G) \leq (n-2\sqrt{n})/4 + O(\log n)$, and that this bound is best possible (up to the $O(\log n)$ term).

2 Result

Our aim is to provide an upper bound on $td(G)$ for a graph G of order n with minimum degree two. It is not too difficult to show that $td(G) \leq n/4$ for such graphs. But one can do better.

We will need the following result.

Lemma 1 *[1] (a) For the cycle C_n with $n \geq 4$, $td(C_n) = 0$ if $4|n$ and 1 otherwise.*
(b) For the path P_n on $n \geq 4$ vertices, $td(P_n) = 1$ if $4|n$ and 2 otherwise.

Before proceeding further, we introduce some additional notation. We define a vertex as ***small*** if it has degree 2, and ***large*** if it has degree more than 2. Given a 2-coloring of the vertices of a graph, we call an edge incident with two vertices of the same color a ***monochromatic edge***.

We say that a vertex v "***needs its own color***" if all its neighbors have the color opposite to v and therefore v needs to be joined to a vertex having its own color. We say that v "***needs the opposite color***" if all its neighbors have the same color as v and therefore v needs to be joined to a vertex having the opposite color to itself.

We consider first the case that the small vertices and large vertices both form independent sets.

Lemma 2 *Let G be a bipartite graph of order n and minimum degree at least 2, with one partite set containing all the small vertices and the other partite set all the large vertices. If there are s small vertices, and ℓ_i vertices of degree i for $i \geq 3$, then,*

$$td(G) \leq \frac{n}{4} - M$$

where

$$M = \frac{1}{16}\ell_3 + \sum_{i\geq 4} \ell_i \left(\frac{1}{4} - 2^{-\lceil i/2\rceil - 1}\right) + \max_{j\geq 3} \frac{s - (L_3 - L_j)(j-1)}{4L_j} - 1$$

and $L_j = \sum_{i\geq j} \ell_i$.

PROOF. Define a (randomized) 2-coloring of the vertices of G as follows. Let $\tilde{G}$ be the graph (possibly with multiple edges) with the small vertices contracted out. Start with a 2-coloring of the vertices of $\tilde{G}$ that minimizes the number of monochromatic edges. This colors the large vertices of G. For each small vertex of G whose two neighbors have the same color, color it with the color opposite it (so it needs its own color). Color the remaining small vertices randomly.

We note that for each vertex of $\tilde{G}$, the majority of its neighbors have the opposite color, since otherwise one can recolor it to produce a 2-coloring with fewer monochromatic edges.

Claim 3
(a) If v is a large vertex incident to no monochromatic edge in $\tilde{G}$, then $\Pr($ *v needs its own color* $) = \Pr($ *v needs the opposite color* $) = 2^{-\deg v}$.

(b) If v is a large vertex incident to a monochromatic edge in $\tilde{G}$, then
(i) $\Pr($ *v needs its own color* $) \leq 2^{-\lceil \deg v/2\rceil}$, *and*
(ii) $\Pr($ *v needs the opposite color* $) = 0$.

(c) A small vertex cannot need the opposite color, and the number of small vertices that need their own color is at most

$$\alpha = \frac{s}{2} - \max_{j\geq 3} \frac{s - (L_3 - L_j)(j-1)}{2L_j}.$$

PROOF. (a) The vertex v is incident with $\deg v$ small vertices. Each of these is colored at random, and so the probability its neighborhood is monochromatic of any particular color is $2^{-\deg v}$.

(b) The vertex v has a small neighbor of the opposite color and so does not need the opposite color. It was incident with at least $\lceil \deg v/2 \rceil$ edges in $\tilde{G}$ that are not monochromatic. So the probability that v needs its own color is at most $2^{-\lceil \deg v/2 \rceil}$.

(c) By the coloring of small vertices, if a small vertex v has a monochromatic neighborhood, then it receives the color opposite to its neighbors, and so needs only its own color.

We will now bound the number of monochromatic edges in $\tilde{G}$ by considering how many monochromatic edges there would be in a random coloring.

For $j \geq 3$, consider a 2-coloring $\mathcal{C}_j$ of $\tilde{G}$ obtained as follows. Take a random balanced coloring of the vertices $\mathcal{L}_j$ of degree at least j, and a random coloring of the remaining vertices. Let e be an edge of $\tilde{G}$ with both ends in $\mathcal{L}_j$. Then if L_j is even, the probability that e is monochromatic is $(L_j/2-1)/(L_j-1)$. (Fix the one end of e; then of the $L_j - 1$ possibilities for its other end, $L_j/2 - 1$ have the same color.) Similarly, if L_j is odd, then the probability that e is monochromatic is at most $(L_j - 1)/(2L_j)$. The probability for L_j odd is the bigger.

On the other hand, at most $X_j = (L_3 - L_j)(j-1)$ edges do not have both ends in $\mathcal{L}_j$. Each of these edges has a 50–50 chance of being monochromatic. It follows that the expected number of monochromatic edges in $\mathcal{C}_j$ is at most

$$\frac{(s - X_j)(L_j - 1)}{2L_j} + \frac{X_j}{2}.$$

Since we actually chose the coloring of $\tilde{G}$ with the fewest monochromatic edges, our coloring has at most that many monochromatic edges, which simplifies to

$$\frac{s}{2} - \frac{s - (L_3 - L_j)(j-1)}{2L_j}.$$

Thus the number of small vertices that need their own color is at most α, as required. $\square$

We now add edges to G so that the given 2-coloring of the vertices has no vertex with a monochromatic neighborhood. More precisely, we add edges to pair up the vertices needing their own color and we add single edges for each (large) vertex needing the opposite color. Let N_1 denote the expected number of vertices that need their own color, and N_2 the expected number that need their opposite color. It follows that the expected number of edges added is at most

$$N_1/2 + N_2 + 1.$$

Let ℓ_i^a be the number of large vertices of degree i incident with no monochromatic edge in $\tilde{G}$. It follows from Claim 3 that

$$N_1 \leq \alpha + \sum_{i \geq 3} \ell_i^a 2^{-i} + \sum_{i \geq 3} (\ell_i - \ell_i^a) 2^{-\lceil i/2 \rceil},$$

and

$$N_2 \leq \sum_{i \geq 3} \ell_i^a 2^{-i}.$$

By inspection it follows that the expression $N_1/2 + N_2 + 1$ is maximized as a function of the ℓ_i^a if $\ell_3^a = \ell_3$ and $\ell_i^a = 0$ for $i \geq 4$, so that the expected number of edges added is at most

$$\frac{\alpha}{2} + \frac{3}{16}\ell_3 + \sum_{i \geq 4} \ell_i 2^{-\lceil i/2 \rceil - 1} + 1,$$

which equals the desired bound since $s + \sum_{i \geq 3} \ell_i = n$. □

As a consequence of Lemma 2, we have the following result.

Lemma 4 *Let G be a bipartite graph on n vertices with minimum degree 2 such that the small vertices form one partite set and the large vertices the other partite set. Then*

$$td(G) \leq \frac{n - 2\sqrt{n}}{4} + c \log n$$

for some constant c, and this result is best possible.

PROOF. The result follows by arithmetic from Lemma 2. Let

$$M_1 = \frac{1}{16}\ell_3 + \sum_{i \geq 4} \ell_i \left(\frac{1}{4} - 2^{-\lceil i/2 \rceil - 1} \right)$$

and

$$M_2 = \max_{j \geq 3} \frac{s - (L_3 - L_j)(j-1)}{4L_j}.$$

Define L_A as the number of large vertices of degree at most $\lceil 2\log n \rceil$ and L_B as the remaining large vertices. Thus $L_B = L_{\lceil 2\log n \rceil + 1}$.

The total contribution of the L_B vertices to M_1 is:

$$\frac{L_B}{4} - \frac{1}{2}\left(\sum_{i \geq \lceil 2\log n \rceil + 1} \ell_i \cdot 2^{-\lceil i/2 \rceil} \right).$$

For $i > 2\log n$, we have $2^{-\lceil i/2 \rceil} < 1/n$. Thus, since $L_B < n$, the expression in parentheses is at most 1. Hence the total contribution of the L_B vertices to M_1 is at least $L_B/4 - 1/2$. The L_A vertices each contribute at least $1/16$ to M_1. So

$$M_1 \geq \frac{L_A}{16} + \frac{L_B}{4} - \frac{1}{2}.$$

Now M_2 is at least the value for $j = \lceil 2\log n \rceil + 1$, and $s = n - L_A - L_B$, so that

$$M_2 \geq \max\left(\frac{n - L_A - L_B - L_A\lceil 2\log n \rceil}{4L_B}, 0 \right).$$

Hence letting $x = L_B$ and $y = L_A$, we wish to minimize $f(x,y) - 3/2$ where

$$f(x,y) = \frac{y}{16} + \frac{x}{4} + \max\left(\frac{n - x - y - y\lceil 2\log n \rceil}{4x}, 0 \right).$$

If $f(x,y) \geq \sqrt{n}/2$, then $M = M_1 + M_2 - 1 \geq \sqrt{n}/2 - 3/2$, and so, by Lemma 2, $td(G) \leq (n - 2\sqrt{n})/4 + 3/2$. Hence we may assume that $f(x,y) < \sqrt{n}/2$ for otherwise the desired result holds. Since $f(x,y) \geq y/16 + x/4$, it follows that $y < 8\sqrt{n}$ and $x < 2\sqrt{n}$. Thus (since $y \geq 0$ and $n \geq 4$), for this range of x and y,

$$f(x,y) \geq \frac{x}{4} + \frac{n - 10\sqrt{n} - 8\sqrt{n}\,\lceil 2\log n \rceil}{4x} \geq \frac{x}{4} + \frac{n - 23\sqrt{n}\log n}{4x}.$$

The function on the far right hand side of the above inequality chain is minimized when $x = \sqrt{n - 23\sqrt{n}\log n}$, and so

$$f(x,y) \geq \frac{1}{2}\sqrt{n - 23\sqrt{n}\log n}.$$

Since by elementary algebra $\sqrt{n-\beta} \geq \sqrt{n} - \beta/\sqrt{n}$ for β sufficiently small, it follows that

$$f(x,y) \geq \frac{1}{2}(\sqrt{n} - 23\log n),$$

and so $f(x,y) \geq \sqrt{n}/2 - c\log n$ for some constant c. Thus, $M = M_1 + M_2 - 1 \geq f(x,y) - 3/2 \geq \sqrt{n}/2 - c\log n - 3/2 \geq \sqrt{n}/2 - (c+1)\log n$. The desired result now follows from Lemma 2. □

That this result is best possible is shown by considering the following graph. Start with the complete graph on $2k$ vertices, duplicate each edge, and then subdivide each edge. The resultant graph G^* has order $n = 4k^2$. The best 2-coloring involves a balanced as possible coloring of the large vertices (i.e., coloring k large vertices with one color and k with the other color), and joining each small vertex needing its own color to another one in the same predicament. Thus,

$$td(G^*) = 2\binom{k}{2} = \frac{n - 2\sqrt{n}}{4}. \qquad \square$$

Before presenting the full result, we introduce some further notation. Let G be a graph with minimum degree at least 2. For $k \geq 2$, we define a ***segment*** of order k as a path $v_1, v_2, \ldots, v_k$ consisting of small vertices, such that v_1 and v_k are either adjacent to different large vertices, or adjacent to the same large vertex which has degree at least 4. For $k \geq 4$, we define a ***lollipop*** of order k as a path $v_1, v_2, \ldots, v_k$, such that for some j with $2 \leq j \leq k-2$, all vertices bar v_j are small, v_j has degree 3 and is adjacent to v_k, and the other neighbor of v_1 is large. These definitions ensure that the removal of a segment or lollipop does not create a vertex with degree less than 2.

We note that: *in G where the large vertices form an independent set but the small vertices do not, then G contains either a segment or a lollipop, or a component that is a cycle.* For, consider two adjacent small vertices. If they do not lie in a segment, nor in a component that is a cycle, then they lie in a cycle that contains exactly one large vertex w, and w has degree 3. All the neighbors of w are small; take the cycle, add the third neighbor of w and continue until one reaches a large vertex. This creates a lollipop.

We define a ***td-optimal* 2*-coloring*** of G as a 2-coloring of the vertices of G in which no vertex has a monochromatic neighborhood when some set of $td(G)$ edges is added to G.

Theorem 5 *For a graph G on $n \geq 4$ vertices with minimum degree at least 2,*

$$td(G) \leq \frac{n - 2\sqrt{n}}{4} + c \log n$$

for some constant c.

PROOF. We may assume that G is edge-minimal with respect to the property of having minimum degree 2. Hence the large vertices of G, if any, form an independent set.

We proceed by induction on the order n of G. By choosing the constant c sufficiently large, the bound can be made true for $n \leq 27$.

Let $n \geq 28$ and assume for all graphs G' of order n', where $n' < n$, with minimum degree at least 2, that $td(G')$ satisfies the desired upper bound. Let G be a graph of order n with minimum degree at least 2.

We proceed further with the following observation.

Induction Observation. *Suppose G' is an induced subgraph of G of order n' where $4 \leq n' < n$ with minimum degree at least 2. If $td(G) \leq td(G')$ or if $n' \leq n - 5$ and $td(G) \leq td(G') + 1$, then $td(G)$ satisfies the desired upper bound.*

PROOF. Applying the inductive hypothesis to G', $td(G')$ satisfies the desired upper bound. If $td(G) \leq td(G')$, then clearly $td(G)$ satisfies the desired upper bound. So assume $n' = n - k$, where $5 \leq k \leq n - 4$, and $td(G) \leq td(G') + 1$ Then, $td(G) \leq td(G') + 1 \leq (n' - 2\sqrt{n'}\,)/4 + c \log n' + 1$, and so

$$td(G) \leq \frac{n - 2\sqrt{n}}{4} + c \log n + \frac{2\sqrt{n} - 2\sqrt{n-k} + 4 - k}{4} + c \log \left(\frac{n-k}{n} \right).$$

For the range of k, the function $2\sqrt{n-k} + k$ is increasing with k and therefore attains its minimum when $k = 5$. Hence, $2\sqrt{n} - 2\sqrt{n-k} + 4 - k \leq 2\sqrt{n} - 2\sqrt{n-5} - 1 \leq 0$ for $n \geq 28$. Since $\log((n-k)/n) < 0$, it follows that for $n \geq 28$, $td(G) \leq (n - 2\sqrt{n}\,)/4 + c \log n$. The desired result follows. $\square$

Case 1: *G has cycle components each of order at least* 4. The result then follows readily by deleting a cycle component, applying induction to the resulting graph, and then using Lemma 1.

Case 2: *G has (at least) two triangle components.* Let G' be obtained from G by deleting two triangle components. Then, $td(G) \le td(G') + 1$ and so, by the Induction Observation, $td(G)$ satisfies the desired upper bound.

Case 3: *G has a lollipop.* Let L be a lollipop of minimum order, say k. Then $G' = G - L$ has minimum degree at least 2. If $G' = K_3$, then G is obtained by adding an edge between two disjoint copies of K_3 and subdividing this edge $k - 3$ times, and so $td(G) = 0$ if k is odd and $td(G) = 1$ if k is even. Hence we may assume that G' has order at least 4. If $4 \le k \le 5$, any td-optimal 2-coloring of G' can be extended to a td-optimal 2-coloring of G without additional edges, and so $td(G) \le td(G')$. If $k \ge 6$, then we can extend any td-optimal 2-coloring of G' to a td-optimal 2-coloring of G by adding at most one edge, and so $td(G) \le td(G') + 1$. In any event, by the Induction Observation, $td(G)$ satisfies the desired upper bound.

Case 4: *G has a segment that is not of order* 3. Let S be a segment of minimum order, say k. Then $G' = G - S$ has minimum degree at least 2. If $G' = K_3$, then by the edge-minimality of G and the choice of S, it follows that G is the bow-tie $K_1 + 2K_2$, for which $td(G) = 1$. So assume that G' has order at least 4. If k is even, then any td-optimal 2-coloring of G' can be extended to a td-optimal 2-coloring of G without additional edges, and so $td(G) \le td(G')$. If k is odd (and therefore at least 5) any td-optimal 2-coloring of G' can be extended to a td-optimal 2-coloring of G by adding at most one edge, and so $td(G) \le td(G') + 1$. Thus, by the Induction Observation, $td(G)$ satisfies the desired upper bound.

Case 5: *All segments have order 3, and there are at least 2 of them.* Let v_1, v_2, v_3 be a segment of G. If v_1 and v_2 are adjacent to the same large vertex, then $G' = G - \{v_1, v_2, v_3\}$ has minimum degree at least 2 and order at least 4, and any td-optimal 2-coloring of G' can be extended to a td-optimal 2-coloring of G without adding any new edges. Thus, $td(G) \le td(G')$ and the result follows from the Induction Observation. Hence we may assume that the end-vertices of any segment are not adjacent to the same large vertex.

Let G' be the graph obtained from G by deleting the six small vertices on two segments. Since $n \ge 28$, we may certainly assume that G' has order at least 4. Any td-optimal 2-coloring of G' can be extended to a td-optimal 2-coloring of G by adding at most one edge. So, if G' has minimum degree at least 2, by the Induction Observation, $td(G)$ satisfies the desired upper bound.

Suppose that G' has a vertex of degree 1. Let v_1, v_2, v_3 and u_1, u_2, u_3 be the two deleted segments of G. We may assume v_1 and u_1 are adjacent to the same large vertex v of degree 3. Let w denote the large vertex adjacent to v_3. Let H be the graph obtained from G' by adding an edge between v and w. Then, H has minimum degree at least 2 and order at least 4. Any td-optimal 2-coloring of H can be extended to a td-optimal 2-coloring of G by adding at most one edge.

Case 6: *G has no lollipop, at most one segment, and if one segment the segment has order 3, and at most one triangle component.* If G has no segment and no triangle component, then the desired result follows from Lemma 4. On the other hand, if G has a segment or a triangle component, then let G' be the graph obtained from G by contracting out two vertices of the segment, if the segment exists, and removing the triangle component, if it exists. Then, G' is a bipartite graph with minimum degree 2 such that the small vertices form one partite set and the large vertices the other partite set. If G has either a segment or a triangle component (but not both), then G' has order $n' \leq n-2$ and any td-optimal 2-coloring of G' can be extended to a td-optimal 2-coloring of G by adding at most one edge, and so $td(G) \leq td(G')+1$. Thus, by Lemma 4,

$$td(G) \leq \frac{n-2\sqrt{n-2}}{4} + c' \log(n-2) + \frac{1}{2}$$

for some constant c'. The desired result now follows readily. Finally, if G has both a segment and a triangle component, then G' has order $n' = n-5$ and any td-optimal 2-coloring of G' can be extended to a td-optimal 2-coloring of G by adding at most two edges, and so $td(G) \leq td(G')+2$. Once again, the desired result follows readily. This completes the cases.

That the result is best possible may be seen by considering the graph G^* described in the proof of Lemma 4. $\square$

3 Open Question

If we turn to larger minimum degrees, consider the problem of 2-coloring the vertices to minimize the number of monochromatic neighborhoods. Trivial random coloring shows that this can be done so that at most $n2^{1-\delta}$ vertices have monochromatic neighbors and this is near best possible. But can one always find a 2-coloring in which no vertex has a monochromatic neighborhood by adding at most $n2^{-\delta}$ edges?

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